TECHNICAL & CODING

100 PYQ-Style Questions — Practice Set 3

Placement & Competitive Exam Practice

Included	100 Distinct PYQ-Style Questions • Practice Set 3
Useful For	TCS NQT • Cognizant • Infosys • Wipro • Accenture • Bosch • Campus Placements
Topics Covered	C • C++ • Java • Python • Data Structures • DBMS & SQL • JavaScript • OS • Networks • OOPs
Extra	Answer Key on Each Page • Practice Set 3 • 1 Page per Language

Structure: 10 Core Programming Languages & Technical Domains. Set 3 contains 100 unique questions. Each language contains exactly 10 authentic MCQs with an Answer Key box at the bottom of the page.

TECHNICAL ASSESSMENT • C PROGRAMMING

Practice Set 3 • Section 1 of 10 • 10 Questions with Answer Key

Q1. What does placing the 'const' keyword before a variable declaration mean in C?

A) Variable is read-only and its value cannot be modified after initialization
B) Stored in ROM
C) Variable is global
D) Compiler allocates on stack

Q2. What is the exact evaluated output of printf("%d", 10 % 3) in C?

A) 3
B) 1 (remainder)
C) 0
D) 3.33

Q3. Which standard library function is invoked to release dynamically allocated heap memory in C?

A) delete()
B) free()
C) dealloc()
D) clear()

Q4. What is the legal index of the last element in a C array declared as int arr[N]?

A) N
B) N - 1
C) N + 1
D) 0

Q5. Which bitwise operator performs the Bitwise XOR operation in C?

A) &
B) |
C) ^
D) ~

Q6. What does the 'typedef' keyword achieve in C programming?

A) Defines a new fundamental data type
B) Creates a convenient alias / synonym for an existing data type
C) Allocates static memory
D) Declares a macro

Q7. Which keyword is used to group variables of different data types under a single user-defined record in C?

A) class
B) struct
C) union
D) record

Q8. What does the macro NULL evaluate to in boolean condition checks in C?

A) True (1)
B) False (0 / null pointer constant)
C) -1
D) Undefined

Q9. What causes a Segmentation Fault runtime crash in a C program?

A) A compilation syntax error
B) Attempting to read or write to an invalid or unauthorized virtual memory address
C) Division by zero
D) Missing return statement

Q10. What is the role of the 'break' statement inside a switch-case block in C?

A) Terminates program execution
B) Jumps execution immediately out of the switch block to prevent fall-through
C) Skips to default case
D) Resets variable state

ANSWER KEY & SOLUTIONS — C PROGRAMMING (SET 3)

Q1: A • Q2: B • Q3: B • Q4: B • Q5: C

Q6: B • Q7: B • Q8: B • Q9: B • Q10: B

TECHNICAL ASSESSMENT • C++ PROGRAMMING & OOPS

Practice Set 3 • Section 2 of 10 • 10 Questions with Answer Key

Q1. Which keyword dynamically allocates memory for an object on the heap in C++?

A) malloc
B) new
C) alloc
D) create

Q2. Which operator must be used to deallocate dynamic memory allocated via new[] for arrays in C++?

A) free()
B) delete[]
C) delete
D) remove[]

Q3. Which C++ STL container provides amortized O(1) insertion at the back and fast random index access?

A) std::list
B) std::vector
C) std::set
D) std::map

Q4. What is the primary benefit of declaring a function as 'inline' in C++?

A) Enables recursion
B) Reduces runtime function call overhead by expanding function body at the call site
C) Restricts access to file
D) Multi-threads execution

Q5. Which algorithm header function returns the smaller of two values in C++?

A) std::less()
B) std::min()
C) std::lower()
D) std::minimum()

Q6. Which access specifier limits member accessibility strictly to the declaring class and its derived subclasses?

A) private
B) protected
C) public
D) friend

Q7. What special requirement applies to a pure virtual destructor in C++?

A) It must have an out-of-class definition/body because derived destructors call it
B) It is illegal in C++
C) Must be private
D) Cannot be overridden

Q8. What is the syntax for publicly inheriting class Derived from class Base in C++?

A) class Derived extends Base
B) class Derived : public Base
C) class Derived implements Base
D) class Derived inherits Base

Q9. Which STL associative container stores unique sorted keys using an internal Red-Black Tree in C++?

A) std::unordered_set
B) std::set
C) std::vector
D) std::deque

Q10. What is nullptr in modern C++11 compared to NULL?

A) Integer 0
B) Type-safe null pointer literal of type std::nullptr_t
C) Void pointer macro
D) Boolean value

ANSWER KEY & SOLUTIONS — C++ PROGRAMMING & OOPS (SET 3)

Q1: B • Q2: B • Q3: B • Q4: B • Q5: B

Q6: B • Q7: A • Q8: B • Q9: B • Q10: B

TECHNICAL ASSESSMENT • JAVA CORE & OOPS

Practice Set 3 • Section 3 of 10 • 10 Questions with Answer Key

Q1. What is the default package-level visibility of a class member in Java if no access modifier is specified?

A) public
B) package-private (default access within same package)
C) protected
D) private

Q2. Which keyword is used to explicitly instantiate and trigger an exception in Java?

A) throws
B) throw
C) raise
D) dispatch

Q3. What is the return value of Math.floor(3.8) in Java?

A) 4.0
B) 3.0
C) 3
D) 3.5

Q4. Which root interface does HashMap implement in Java's collections hierarchy?

A) java.util.Collection
B) java.util.Map
C) java.util.List
D) java.util.Set

Q5. What thread lifecycle state corresponds to a thread waiting to acquire an intrinsic monitor lock in Java?

A) RUNNABLE
B) BLOCKED
C) WAITING
D) TIMED_WAITING

Q6. Can a static method in Java access an instance variable directly without creating an object reference?

A) Yes, always
B) No, because static methods exist at class level and lack an implicit 'this' reference
C) Only if protected
D) Only in main()

Q7. Which Java keyword is applied to methods or blocks to enforce mutual exclusion across concurrent threads?

A) volatile
B) synchronized
C) atomic
D) transient

Q8. What is the exact evaluated output of the expression 10 + 20 + "Java" in Java?

A) "1020Java"
B) "30Java" (addition occurs before string concatenation)
C) "Java30"
D) Compilation Error

Q9. What is the role of the 'transient' keyword modifier in Java class fields?

A) Signals that the field should be skipped and NOT included during object serialization
B) Makes field immutable
C) Marks field as thread-safe
D) Allocates in stack

Q10. Which garbage collection algorithm serves as the default low-pause collector in modern HotSpot JVMs?

A) G1 GC (Garbage-First Collector)
B) Serial GC only
C) Reference counting only
D) Stop-the-world single thread

ANSWER KEY & SOLUTIONS — JAVA CORE & OOPS (SET 3)

Q1: B • Q2: B • Q3: B • Q4: B • Q5: B

Q6: B • Q7: B • Q8: B • Q9: A • Q10: A

TECHNICAL ASSESSMENT • PYTHON PROGRAMMING

Practice Set 3 • Section 4 of 10 • 10 Questions with Answer Key

Q1. What is the evaluated output of the floor division expression 5 // 2 in Python 3?

A) 2.5
B) 2
C) 3
D) 2.0

Q2. Which file access mode string opens a file for both reading and writing without truncating in Python?

A) 'r'
B) 'r+'
C) 'w'
D) 'a'

Q3. Which built-in function returns the total count of elements contained in a Python sequence?

A) size()
B) count()
C) len()
D) length()

Q4. What does the 'is' identity operator check between two objects in Python?

A) Value equality (==)
B) Whether both variables reference the exact same memory object (id(a) == id(b))
C) Subclass relation
D) Type matching

Q5. What is the evaluated list produced by list(range(3)) in Python?

A) [1, 2, 3]
B) [0, 1, 2]
C) [0, 1, 2, 3]
D) (0, 1, 2)

Q6. Which loop control statement exits immediately from the innermost enclosing loop in Python?

A) continue
B) break
C) exit
D) pass

Q7. How do you insert or update a key-value mapping in an existing Python dictionary d?

A) d.add(key, val)
B) d[key] = val
C) d.append(key, val)
D) d.insert(key, val)

Q8. What is produced by list(enumerate(['x', 'y'])) in Python?

A) [(0, 'x'), (1, 'y')]
B) ['x', 'y']
C) [0, 1]
D) {'x': 0, 'y': 1}

Q9. Which method removes and returns the last element from a Python list?

A) remove()
B) pop()
C) delete()
D) discard()

Q10. What defines a Generator function in Python?

A) A function that generates random numbers
B) A function containing the 'yield' keyword that returns an iterator yielding values lazily
C) Class constructor
D) Lambda expression

ANSWER KEY & SOLUTIONS — PYTHON PROGRAMMING (SET 3)

Q1: B • Q2: B • Q3: C • Q4: B • Q5: B

Q6: B • Q7: B • Q8: A • Q9: B • Q10: B

TECHNICAL ASSESSMENT • DATA STRUCTURES & ALGORITHMS

Practice Set 3 • Section 5 of 10 • 10 Questions with Answer Key

Q1. What is the time complexity of pushing an element onto an array-based stack?

A) O(1)
B) O(N)
C) O(log N)
D) O(N^2)

Q2. Which data structure is the ideal architectural choice for implementing an 'Undo' buffer in text editors?

A) Queue
B) Stack
C) Binary Tree
D) Hash Table

Q3. In a Max-Heap data structure, where is the maximum element always guaranteed to reside?

A) At the root node (index 0)
B) At the leftmost leaf
C) At the rightmost leaf
D) At mid-level

Q4. What is the worst-case time complexity of Binary Search on a sorted array of size N?

A) O(1)
B) O(log N)
C) O(N)
D) O(N log N)

Q5. Which underlying data structure is most efficiently used to implement a Priority Queue?

A) Stack
B) Binary Heap
C) Circular Linked List
D) Hash Set

Q6. What is the auxiliary call-stack space complexity of Depth-First Search (DFS) on a tree of height H?

A) O(H)
B) O(V^2)
C) O(1)
D) O(E)

Q7. Which sorting algorithm recursively divides the input list into halves, sorts them, and merges them?

A) Insertion Sort
B) Merge Sort
C) Bubble Sort
D) Selection Sort

Q8. How many pointer fields does each individual node contain in a standard Doubly Linked List?

A) 1
B) 2 (previous and next pointer)
C) 3
D) 0

Q9. What is the computational time complexity of accessing an element by index in a contiguous array?

A) O(1)
B) O(N)
C) O(log N)
D) O(N^2)

Q10. Which specialized tree data structure is optimal for autocomplete and dictionary prefix queries?

A) Binary Search Tree
B) Trie (Prefix Tree)
C) Red-Black Tree
D) B+ Tree

ANSWER KEY & SOLUTIONS — DATA STRUCTURES & ALGORITHMS (SET 3)

Q1: A • Q2: B • Q3: A • Q4: B • Q5: B

Q6: A • Q7: B • Q8: B • Q9: A • Q10: B

TECHNICAL ASSESSMENT • SQL & DATABASE MANAGEMENT (DBMS)

Practice Set 3 • Section 6 of 10 • 10 Questions with Answer Key

Q1. Which SQL statement is used to insert new rows of data into a database table?

A) ADD ROW
B) INSERT INTO
C) UPDATE
D) CREATE

Q2. Which SQL aggregate function counts the total number of rows matching a given filter condition?

A) SUM()
B) COUNT()
C) TOTAL()
D) ROWS()

Q3. What is a Candidate Key in relational database schema design?

A) A minimal superkey that can uniquely identify records in a table
B) A foreign key reference
C) Any indexed column
D) Primary key duplicate

Q4. What does the SQL BETWEEN operator test when filtering queries?

A) Checks whether an attribute falls within an inclusive numeric or date range
B) Performs string equality
C) Joins two tables
D) Filters NULLs

Q5. What is a Self Join in relational SQL queries?

A) A regular join in which a table is joined with itself based on a common key
B) Merging two databases
C) Cartesian product
D) Union of tables

Q6. Which constraint enforces that all recorded values in a table column remain distinct from one another?

A) NOT NULL
B) UNIQUE
C) CHECK
D) DEFAULT

Q7. What is a Stored Procedure in enterprise database systems?

A) A precompiled collection of SQL statements and logic stored directly on the database engine
B) Database dump file
C) A table view
D) An index tree

Q8. Which database normalization form specifically addresses and resolves Multi-Valued Dependencies?

A) 3NF
B) BCNF
C) 4NF (Fourth Normal Form)
D) 2NF

Q9. Which SQL clause specifies the row-level filtering condition in a standard SELECT query?

A) HAVING
B) WHERE
C) ORDER BY
D) GROUP BY

Q10. What is a Database Trigger in relational database management?

A) Stored procedural code executed automatically when a specified event (INSERT, UPDATE, DELETE) occurs on a table
B) A cron job
C) A login script
D) A commit command

ANSWER KEY & SOLUTIONS — SQL & DATABASE MANAGEMENT (DBMS) (SET 3)

Q1: B • Q2: B • Q3: A • Q4: A • Q5: A

Q6: B • Q7: A • Q8: C • Q9: B • Q10: A

TECHNICAL ASSESSMENT • JAVASCRIPT & WEB TECHNOLOGIES

Practice Set 3 • Section 7 of 10 • 10 Questions with Answer Key

Q1. What is the evaluated output of typeof [] in standard JavaScript?

A) 'array'
B) 'object'
C) 'list'
D) 'undefined'

Q2. Which Array method removes and returns the first element from an array in JavaScript?

A) pop()
B) shift()
C) unshift()
D) slice()

Q3. What is the evaluated result of the string concatenation 2 + "2" in JavaScript?

A) 4
B) "22"
C) NaN
D) TypeError

Q4. Which keyword immediately exits function execution and passes back a computed value?

A) break
B) return
C) stop
D) yield

Q5. What is the primary characteristic of browser localStorage compared to sessionStorage?

A) Persists data across browser sessions with no expiration date
B) Clears data when current tab is closed
C) Automatically sent with HTTP requests
D) Limited to 4KB

Q6. Which Array method concatenates all elements of an array into a single delimited string?

A) concat()
B) join()
C) split()
D) merge()

Q7. What is the boolean evaluation of Boolean(0) in JavaScript?

A) true
B) false
C) null
D) undefined

Q8. What does document.getElementById('unknownId') return if no matching DOM element is found?

A) undefined
B) null
C) false
D) empty HTMLCollection

Q9. Which operator represents the Nullish Coalescing Operator in modern JavaScript (ES2020)?

A) ||
B) ??
C) &&
D) ?:

Q10. What does calling Array.prototype.map() return in JavaScript?

A) Modifies array in place
B) A new array containing the transformed elements
C) A single accumulated primitive
D) A boolean flag

ANSWER KEY & SOLUTIONS — JAVASCRIPT & WEB TECHNOLOGIES (SET 3)

Q1: B • Q2: B • Q3: B • Q4: B • Q5: A

Q6: B • Q7: B • Q8: B • Q9: B • Q10: B

TECHNICAL ASSESSMENT • OPERATING SYSTEMS & LINUX

Practice Set 3 • Section 8 of 10 • 10 Questions with Answer Key

Q1. What is the primary role of an Operating System kernel in computer architecture?

A) Text editor
B) Core software acting as an intermediary managing hardware resources and process execution
C) Compiler
D) Database manager

Q2. What is a Critical Section in concurrent multithreaded programming?

A) A code section accessing shared variables that must not be concurrently executed by multiple threads
B) Deadlock zone
C) Memory crash area
D) Bootloader segment

Q3. Which CPU scheduling algorithm assigns equal fixed time slices (quanta) to processes in a cyclic ring?

A) FCFS
B) Round Robin (RR)
C) Priority Scheduling
D) Shortest Job First

Q4. What is Internal Fragmentation in fixed-partition memory allocation?

A) Unused memory wasted inside an allocated partition because the process requested less than partition size
B) External scatter
C) Swapping delay
D) Cache miss

Q5. Which POSIX system call is executed to terminate the calling process in Unix/Linux?

A) stop()
B) exit()
C) end()
D) kill()

Q6. What constitutes a Context Switch in CPU multitasking?

A) Compiling C code
B) Saving the CPU register state of an active process and loading the state of the next scheduled process
C) RAM defragmentation
D) Changing shell theme

Q7. Which Linux terminal utility is used to list directory contents and file metadata?

A) cd
B) ls
C) mkdir
D) pwd

Q8. What is Starvation in Operating System resource scheduling?

A) Hardware overheating
B) A runnable process waiting indefinitely for CPU or resources because higher priority tasks dominate
C) Out of memory error
D) Hard drive failure

Q9. What is Deadlock in operating system process synchronization?

A) A state where a set of processes are permanently blocked because each holds resources needed by others
B) High CPU usage loop
C) Infinite recursive call
D) Stack overflow

Q10. What is the difference between a Hard Link and a Soft (Symbolic) Link in Linux file systems?

A) Hard link points directly to file inode; Soft link points to file pathname as a pointer
B) Hard link only works for folders
C) Soft link cannot be deleted
D) No difference

ANSWER KEY & SOLUTIONS — OPERATING SYSTEMS & LINUX (SET 3)

Q1: B • Q2: A • Q3: B • Q4: A • Q5: B

Q6: B • Q7: B • Q8: B • Q9: A • Q10: A

TECHNICAL ASSESSMENT • COMPUTER NETWORKS

Practice Set 3 • Section 9 of 10 • 10 Questions with Answer Key

Q1. What is the topmost Layer 7 in the OSI (Open Systems Interconnection) reference model?

A) Presentation Layer
B) Application Layer
C) Transport Layer
D) Session Layer

Q2. Which standard protocol is used for uploading and downloading files between client and server?

A) HTTP
B) FTP (File Transfer Protocol)
C) SMTP
D) DNS

Q3. What is the standard well-known TCP port number for unencrypted HTTP web traffic?

A) 21
B) 80
C) 443
D) 25

Q4. What does IP stand for in the fundamental TCP/IP networking protocol suite?

A) Internet Protocol
B) Internal Packet
C) Interconnect Port
D) Interface Path

Q5. Which network topology connects every computer node directly to a central hub or switch?

A) Bus Topology
B) Star Topology
C) Ring Topology
D) Mesh Topology

Q6. What is the designated IPv4 Loopback Address used for testing local host services?

A) 0.0.0.0
B) 127.0.0.1
C) 192.168.1.1
D) 255.255.255.255

Q7. Which transport protocol does NOT guarantee delivery, sequencing, or error acknowledgment?

A) TCP
B) UDP (User Datagram Protocol)
C) SCTP
D) BGP

Q8. What does LAN stand for in computer networking terminology?

A) Local Area Network
B) Logical Array Node
C) Line Access Network
D) Link Area Node

Q9. Which network device is designed to inspect destination IP addresses and forward packets across different subnets?

A) Hub
B) Router
C) Repeater
D) Bridge

Q10. What is Bandwidth in data communication networks?

A) Physical length of fiber cable
B) The maximum data carrying capacity / transfer rate of a communication channel per second
C) Ping latency in milliseconds
D) Packet drop rate

ANSWER KEY & SOLUTIONS — COMPUTER NETWORKS (SET 3)

Q1: B • Q2: B • Q3: B • Q4: A • Q5: B

Q6: B • Q7: B • Q8: A • Q9: B • Q10: B

TECHNICAL ASSESSMENT • OBJECT-ORIENTED PROGRAMMING (OOPS)

Practice Set 3 • Section 10 of 10 • 10 Questions with Answer Key

Q1. What is Inheritance in object-oriented software engineering?

A) Restricting private access

B) A mechanism allowing a new derived class to inherit fields and methods from an existing base class

C) Dynamic method lookup

D) Garbage collection

Q2. What is an Interface in object-oriented design?

A) A contract specifying abstract method signatures that any implementing class must provide

B) Graphical user interface

C) Database schema

D) A private class

Q3. What is Method Overloading in class design?

A) Overriding base class methods

B) Defining multiple methods within the same class having identical names but differing parameter types/counts

C) Calling recursive functions

D) Destructor calls

Q4. Which OOP pillar allows treating objects of different specialized types through a uniform common base class?

A) Encapsulation

B) Polymorphism

C) Modularity

D) Static Binding

Q5. What is an Abstract Class in object-oriented programming?

A) A class that cannot be directly instantiated and is designed to serve as a base blueprint for subclasses

B) A class with no variables

C) A final class

D) An interface with only static variables

Q6. What is a Destructor in object lifecycle management in C++?

A) Method that allocates memory

B) A special member function automatically executed when an object goes out of scope to release resources

C) Constructor alias

D) Memory defragmenter

Q7. What is Data Hiding in object-oriented encapsulation?

A) Declaring member variables private to prevent direct unauthorized access and mutation from outside

B) Erasing variables

C) File encryption

D) Code minification

Q8. Can static methods be dynamically overridden with polymorphic behavior in Java?

A) Yes, runtime override

B) No, static methods are bonded at compile-time (method hiding, not dynamic overriding)

C) Only if public

D) Only in abstract classes

Q9. What is Aggregation in object relationship modeling?

A) Child object lifetime is strictly tied to parent

B) A weak 'HAS-A' relationship where child object can exist independently of parent object

C) Multiple inheritance

D) Friend function

Q10. What is the purpose of the 'super' keyword in Java object inheritance?

A) References current object

B) Refers directly to the immediate parent (superclass) instance or constructor

C) Marks class as public

D) Replaces interface

ANSWER KEY & SOLUTIONS — OBJECT-ORIENTED PROGRAMMING (OOPS) (SET 3)

Q1: B • Q2: A • Q3: B • Q4: B • Q5: A

Q6: B • Q7: A • Q8: B • Q9: B • Q10: B